

Aguaviva

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ABSTRACT

Aguaviva juxtaposes the spontaneous nature of biology with the predictable properties of digital technology.

A solitary moon jellyfish swims around in a saltwater dome. A small camera tracks its movement and turns it into xy values, expressed on a collar of digital numbers. The shifting position of the jellyfish is mapped to the corresponding digits below, resulting in an ever-changing string of random numbers.

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Arbitrary values generated by computers are considered too predictable for high-end encryption schemes, such as secure Internet traffic and online banking. Hence, more unconventional sources are often used, even paid for. True randomness, for this purpose, is a commodity.

As part of the artwork, the numerical string created by the jellyfish is offered up in real-time to encryption companies to use at their discretion.

The apparatus is designed to extract randomness from this simple yet ancient life form—unaware of the fact that in the arena of random sequencing, its cellular contractions can outperform even the most powerful supercomputer.

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